

1. Create Database

```
CREATE DATABASE Hospital_Management;  
USE Hospital_Management;
```

2. Create Tables

```
CREATE TABLE Doctor (  
    Doctor_ID INT PRIMARY KEY,  
    Doctor_Name VARCHAR(50),  
    Specialization VARCHAR(50),  
    Phone VARCHAR(15)  
);
```

```
CREATE TABLE Patient (  
    Patient_ID INT PRIMARY KEY,  
    Patient_Name VARCHAR(50),  
    Age INT,  
    Gender VARCHAR(10),  
    Phone VARCHAR(15)  
);
```

```
CREATE TABLE Appointment (  
    Appointment_ID INT PRIMARY KEY,  
    Patient_ID INT,  
    Doctor_ID INT,  
    Appointment_Date DATE,  
    FOREIGN KEY (Patient_ID) REFERENCES Patient(Patient_ID),  
    FOREIGN KEY (Doctor_ID) REFERENCES Doctor(Doctor_ID)  
);
```

```
CREATE TABLE Treatment (  
    Treatment_ID INT PRIMARY KEY,  
    Patient_ID INT,  
    Doctor_ID INT,  
    Treatment_Name VARCHAR(100),  
    Treatment_Date DATE,  
    FOREIGN KEY (Patient_ID) REFERENCES Patient(Patient_ID),  
    FOREIGN KEY (Doctor_ID) REFERENCES Doctor(Doctor_ID)  
);
```

```
CREATE TABLE Billing (  
    Bill_ID INT PRIMARY KEY,  
    Patient_ID INT,
```

```
Amount DECIMAL(10,2),
Bill_Date DATE,
Payment_Status VARCHAR(20),
FOREIGN KEY (Patient_ID) REFERENCES Patient(Patient_ID)
);
```

3. Insert Values

```
INSERT INTO Doctor VALUES
(101, 'Dr. Arun', 'Cardiology', '9876543210'),
(102, 'Dr. Priya', 'Neurology', '9876543211'),
(103, 'Dr. Kumar', 'Orthopedics', '9876543212');
```

```
INSERT INTO Patient VALUES
(201, 'Ravi', 35, 'Male', '9000000001'),
(202, 'Anitha', 28, 'Female', '9000000002'),
(203, 'Suresh', 45, 'Male', '9000000003');
```

```
INSERT INTO Appointment VALUES
(301, 201, 101, '2026-10-01'),
(302, 202, 102, '2026-10-02'),
(303, 203, 103, '2026-10-03');
```

```
INSERT INTO Treatment VALUES
(401, 201, 101, 'Heart Checkup', '2026-10-01'),
(402, 202, 102, 'Brain Scan', '2026-10-02'),
(403, 203, 103, 'Bone Treatment', '2026-10-03');
```

```
INSERT INTO Billing VALUES
(501, 201, 5000.00, '2026-10-01', 'Paid'),
(502, 202, 3500.00, '2026-10-02', 'Pending'),
(503, 203, 4500.00, '2026-10-03', 'Paid');
```

4. SQL Queries

Display all patients

```
SELECT * FROM Patient;
```

Display all doctors

```
SELECT * FROM Doctor;
```

Display patient and doctor appointment details

```
SELECT P.Patient_Name, D.Doctor_Name, D.Specialization,  
       A.Appointment_Date  
FROM Appointment A  
JOIN Patient P ON A.Patient_ID = P.Patient_ID  
JOIN Doctor D ON A.Doctor_ID = D.Doctor_ID;
```

Display treatment details

```
SELECT P.Patient_Name, D.Doctor_Name,  
       T.Treatment_Name, T.Treatment_Date  
FROM Treatment T  
JOIN Patient P ON T.Patient_ID = P.Patient_ID  
JOIN Doctor D ON T.Doctor_ID = D.Doctor_ID;
```

Display pending bills

```
SELECT * FROM Billing  
WHERE Payment_Status = 'Pending';
```

Calculate total billing amount

```
SELECT SUM(Amount) AS Total_Amount  
FROM Billing;
```

Find patients above 40 years

```
SELECT * FROM Patient  
WHERE Age > 40;
```

Update payment status

```
UPDATE Billing  
SET Payment_Status = 'Paid'  
WHERE Bill_ID = 502;
```

Delete a patient

```
DELETE FROM Patient  
WHERE Patient_ID = 203;
```

5. Sample Output

Patient_ID Patient_Name Age Gender

201	Ravi	35	Male
202	Anitha	28	Female
203	Suresh	45	Male

Patient_Name Doctor_Name Specialization Appointment_Date

Ravi	Dr. Arun	Cardiology	2026-10-01
Anitha	Dr. Priya	Neurology	2026-10-02
Suresh	Dr. Kumar	Orthopedics	2026-10-03